

🚀 ISRO Exoplanet Pipeline

AI-Enabled Detection from Noisy Light Curves

Select Demo Case

🚀 Kepler-7b — Real NASA Conf... ▾

🚀 Run Pipeline

Pipeline Stages

- S0 Data Quality
- S1 Domain Trust T
- S2 3-Method Denoising
- S3 Blend Correction
- S4 BLS Detection
- S5 RF+CNN+GB Classify
- S6 Uncertainty Aggregation

About

Built for ISRO BAH 2026.

Hierarchical uncertainty propagation with Domain Trust T, three-method denoising, and architecturally diverse ensemble classification.

🪐 AI-Enabled Exoplanet Detection Pipeline

An uncertainty-aware adaptive pipeline designed for AstroSat UVIT photometry. Every detection comes with a calibrated confidence score, full audit trail, and explicit ruling-out of false positives.

🔍 FLAGGED FOR HUMAN REVIEW

Candidate: KEPLER-7b (Real Parameters) | True Label: Planet Transit



📶 Raw Input 🔍 Denoising 🔭 Transit Detection ⚙️ Classification 📊 Uncertainty 📄 Full Report

Complete Explainability Report

Every decision is traceable. Every alternative is explicitly ruled out. This is what gets sent to the astronomer review queue.

Ruling Out Alternatives

- ▼ ☒ Eclipsing Binary

No secondary eclipse at phase ~0.5. Depth ratio 0.000 (threshold: 0.5)
- ▼ ⚠️ Starspot

Starspot probability 49.8%. Transit morphology inconsistent with sinusoidal rotational modulation.
- ▼ ☒ Instrumental Artifact

BLS SNR = 12.0 > 7. Signal present across multiple independent observation windows.
- ▼ ☒ Stellar Blending

No centroid shift detected. Target flux fraction = 0.90. Depth corrected for blending.

Recommended Follow-up

-  Standard follow-up photometry recommended.

Raw Audit Trail (JSON)

```
{
  "candidate_id": "KEPLER-7b (Real Parameters)"
  "true_label": "Planet Transit"
  "final_decision": "FLAGGED FOR HUMAN REVIEW"
  "confidence": 0.135
  "confidence_interval": {
    "lower": 0
    "upper": 0.3779
  }
  "domain_trust_T": 0.9991
  "trust_level": "HIGH"
  "transit_parameters": {
    "period_days": 4.88458
    "depth_percent": 0.6398
    "duration_hours": 2.88
    "bls_snr": 11.97
  }
  "classification": {
    "predicted": "Starspot"
    "probabilities": {
      "Planet Transit": 0.1522
      "Eclipsing Binary": 0.0748
      "Starspot": 0.498
      "Instrumental": 0.275
    }
    "model_agreement": false
    "sigma_classifier": 0.1151
  }
  "uncertainties": {
    "sigma_denoise": 0.1458
    "sigma_blend": 0.1111
    "sigma_BLS": 0.0771
    "sigma_classifier": 0.1151
    "sigma_total": 0.1128343110813926
    "dominant_source": "blend"
  }
  "denoising": {
    "agreement_score": 0.8542
    "method": "ensemble (Savitzky-Golay + Wavelet + Median Filter)"
    "interpretation": "High confidence - three methods agree"
  }
}
```

```
}  
}
```